

# CRYPTOKAVACH

*Cryptocurrency Address Threat Intelligence & Monitoring System*

[ INVESTIGATIVE TOOL // THREAT INTELLIGENCE AND MONITORING SYSTEM ]

## SYSTEM STATS

**50,000+**

Addresses

**17**

Threat Categories

**7**

Coin Types

**Upto 85%**

ML Accuracy

**4**

Core Modules

# What is CryptoKavach?

CryptoKavach is an autonomous cryptocurrency address threat intelligence and monitoring system built for law enforcement and investigative agencies to track, categorize, and monitor suspicious blockchain activity in real-time.



## SCALE

Billions of crypto transactions daily — impossible to track manually



## IDENTIFY

Criminals hide behind anonymous wallet addresses with no identity.



## SPEED

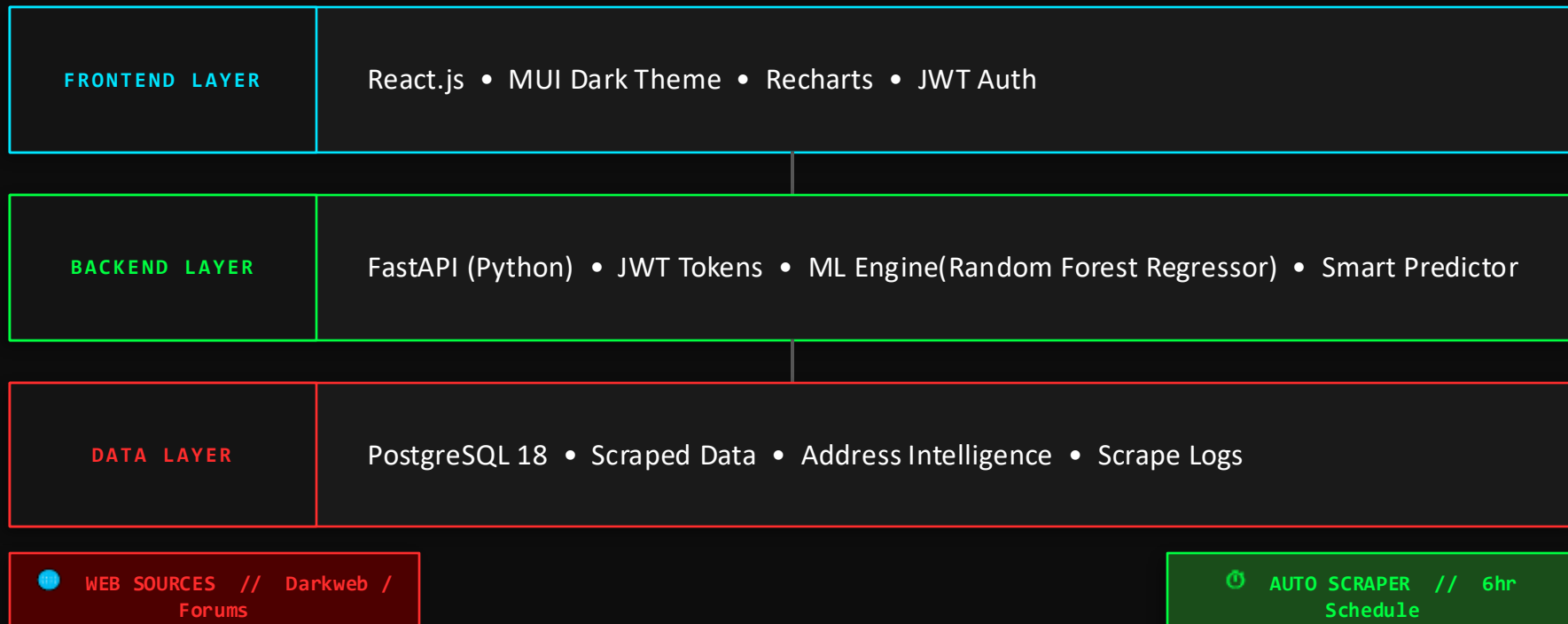
Evidence disappears fast — funds move within minutes of a crime



## CONNECT

Linking addresses to known threat actors requires intelligence

# System Architecture/Technical Approach



# Core Features

## User Auth

JWT login, account lockout after 3 failed attempts, role-based access (Admin/Investigator), strong password enforcement

## Address Search

Full-text search, filter by coin/category/date/source, paginated results, CSV/JSON export, address detail modal

## Auto Scraper

Autonomous scraping every 6 hours from darkweb, forums, news sources etc. Logs all jobs with status tracking

## ML Categorization

Categorize uncategorized addresses with ML prediction. Confidence scoring and accuracy display. The Random Forest Regressor Model shows upto 85% accuracy.

## Dashboard

Live stats cards, donut chart by category, bar charts by coin type and source. Real-time data from PostgreSQL

## Address Tagging

Smart predictor using coin type + source + address patterns for categorizing new addresses.

# Technology Stack

| FRONTEND          | BACKEND          | ML & AI               | DATA & INFRA   |
|-------------------|------------------|-----------------------|----------------|
| React.js 18       | Python 3.13      | Scikit-learn          | PostgreSQL 18  |
| Material UI (MUI) | FastAPI          | Random Forest         | psycopg3       |
| Recharts          | Uvicorn          | TF-IDF Vectorizer     | BeautifulSoup4 |
| Axios             | JWT / bcrypt     | Smart Predictor Rules | Trafilatura    |
| LocalStorage JWT  | Schedule Library | Joblib (model save)   | python-dotenv  |

# System Modules

## Dashboard

- › Total Addresses: 50,000+
- › Categories: 17 threat types
- › Live charts: coin / source
- › Successful scrapes: tracked

## ML Categorization

- › Train model on labeled data
- › Predict uncategorized addresses
- › Review: Save to Database ☒ / Cancel ☐

## Address Search

- › Filter: coin / category / date
- › Show: 100 / 500 / 1000 / All
- › Export: CSV and JSON format
- › Detail modal with full information and the facility to export it in the PDF format

## Blockchain Enrichment

- › Input address
- › Shows the live transaction history of the address entered
- › Risk profile displayed
- › Flow graph displayed